Experimental Quantum Fingerprinting

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Juan Miguel Arrazola

Institute for Quantum Computing
University of Waterloo

Feihu Xu*

Dept. of Electrical and Computer Engineering
University of Toronto

**Currently, at MIT*

QCrypt 2015, Tokyo













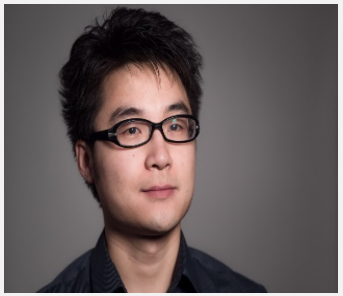
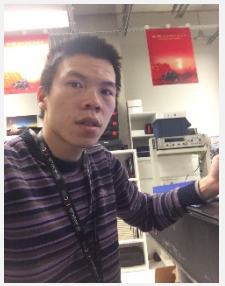
People

- **Theory: Institute for Quantum Computing**
  - Norbert Lütkenhaus
  - Juan M. Arrazola*
  - Pablo Palacios-Avila
  - Shihan Sajeed






- **Experiment: U. of Toronto**
  - Hoi-Kwong Lo
  - Feihu Xu*
  - Kejin Wei

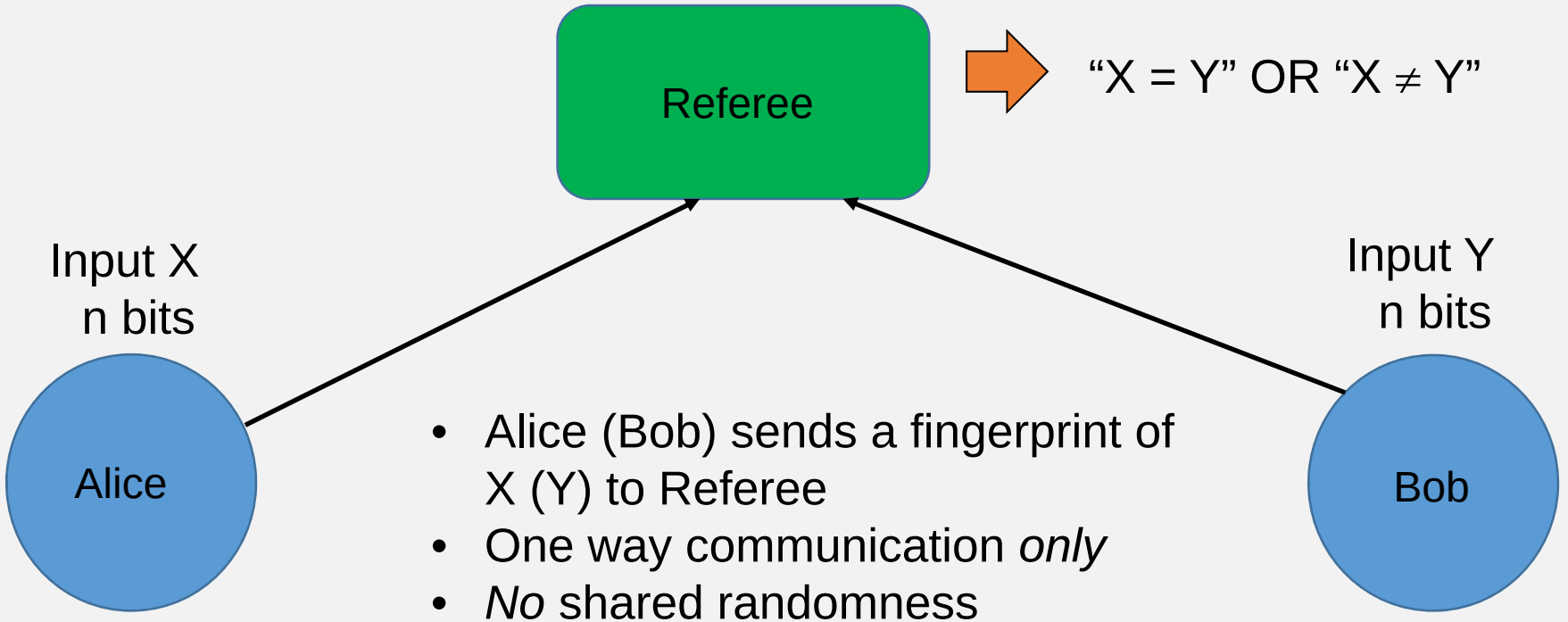




- **ECC: U. of British Columbia & U. of Hong Kong**
  - Chen Feng
  - Wenyuan Wang




* Equal contribution to this work

Fingerprinting model
(simultaneous message passing)



- Alice (Bob) sends a fingerprint of X (Y) to Referee
- One way communication *only*
- *No* shared randomness



Given two n-bit strings X and Y, what is the smallest amount of information that must be transmitted to the referee?

A. C.-C. Yao, in Proceedings of the 11th Annual ACM Symposium on Theory of Computing, pp. 209-213, 1979.

Why quantum fingerprinting?

Proven classical bound*:

$$\Omega(\sqrt{n})$$

Proven quantum bound**:

$$O(\log_2 n)$$

Quantum systems of dimension n can transmit at most $O(\log_2 n)$ qubits of information.

An *exponential* gap between classical and quantum!

* A. Ambainis, Algorithmica **16**, 298 (1996).

** H. Buhrman, R. Cleve, J. Watrous, R. Wolf , *Phys. Rev. Lett.*, *87*, 167902 (2001).

Can we demonstrate the quantum advantage?

Need to prepare the states: $|x\rangle := \frac{1}{\sqrt{m}} \sum_{i=1}^{m} (-1)^{E(x)_i} |i\rangle \quad m > n$

1) Qubits

- ➔ Need to generate $log_2(n)$ **highly entangled** qubits
- ➔ Need to perform complex operations on these qubits
- ➔ Limit previous experiments to a **single qubit**
  [Horn *et al., PRL* **95**, 150502 (2005)]



2) Qudits Single photon across m modes, e.g. time-bin, spatial, OAM

- ➔ Hard to prepare for very large m
- ➔ Susceptible to losses and imperfections



3) Coherent States: Choose laser pulses (coherent states) + time-bin modes



$$|\alpha, x\rangle := \bigotimes_{i=1}^{m} \left| (-1)^{E(x)_i} \frac{\alpha}{\sqrt{m}} \right\rangle_i$$

[Arrazola, Lütkenhaus, *Phys. Rev. A,* **89**, 062305, (2014)]

Transmitted Information

States of a single photon across m modes span a Hilbert space of dimension m, which is mathematically equivalent to $\mathbf{O(\log_2 m)}$ **qubits.**

$$|\alpha, x\rangle := \bigotimes_{i=1}^{m} \left| (-1)^{E(x)_i} \frac{\alpha}{\sqrt{m}} \right\rangle_i \qquad \mu_{total} = |\alpha|^2$$

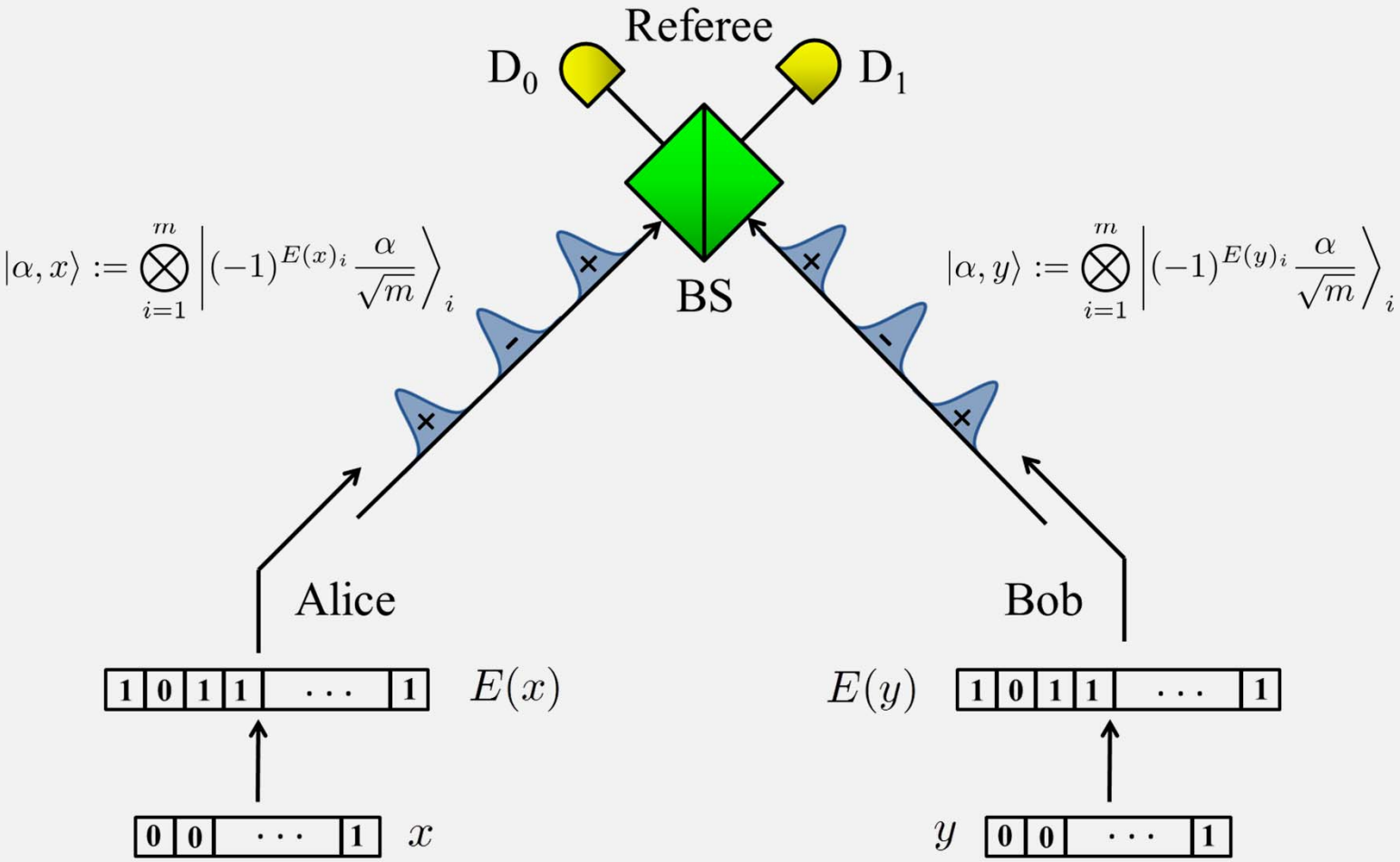
A coherent state with mean photon number $\mu = |\alpha|^2$ across m modes spans a Hilbert space mathematically equivalent to $\mathbf{O(\mu \log_2 m)}$ **qubits.**

By restricting the total mean photon number to a small constant, independent of m, we restrict the capability of these states to transmit information.

Number of modes is linear in the input size.

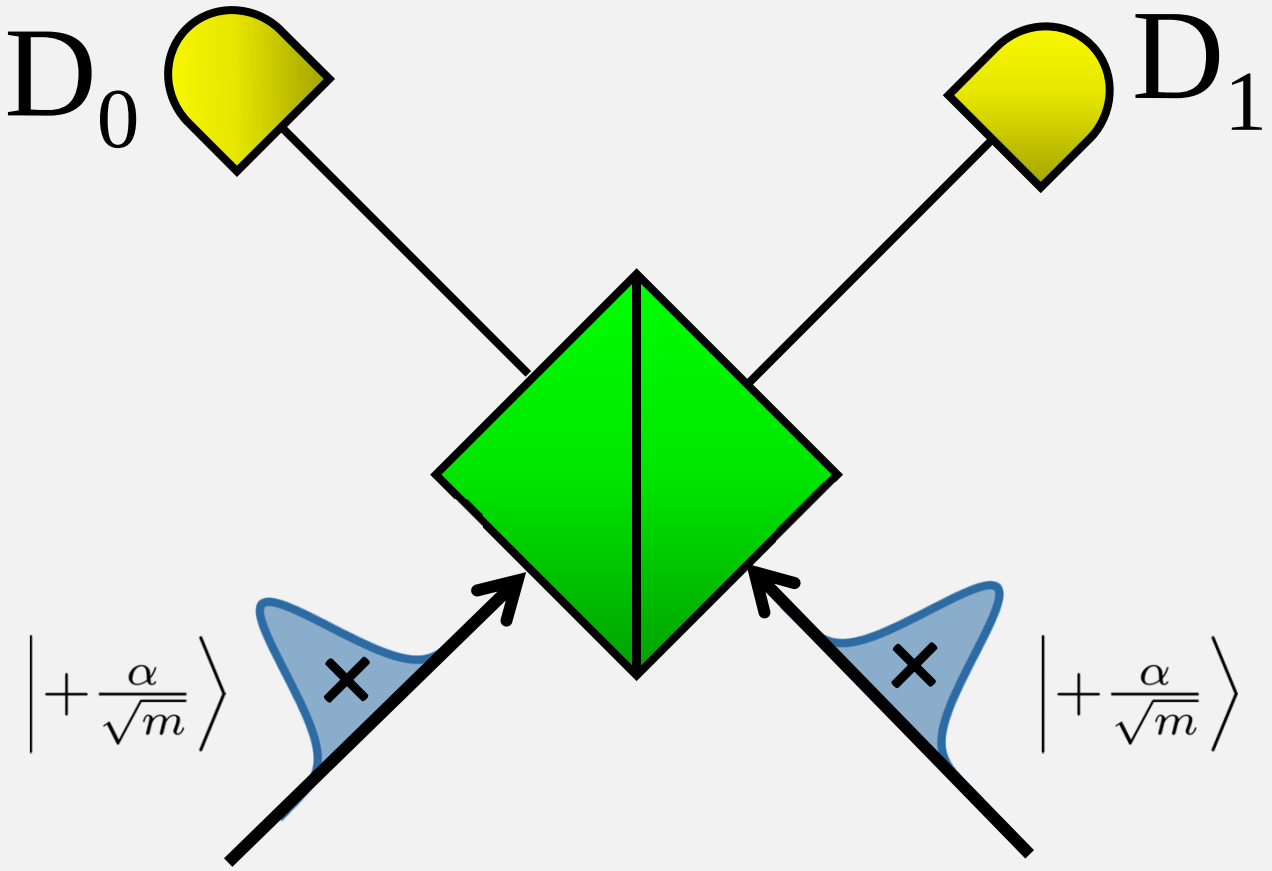
Number of photons is small and independent of input size.

A practical optical scheme

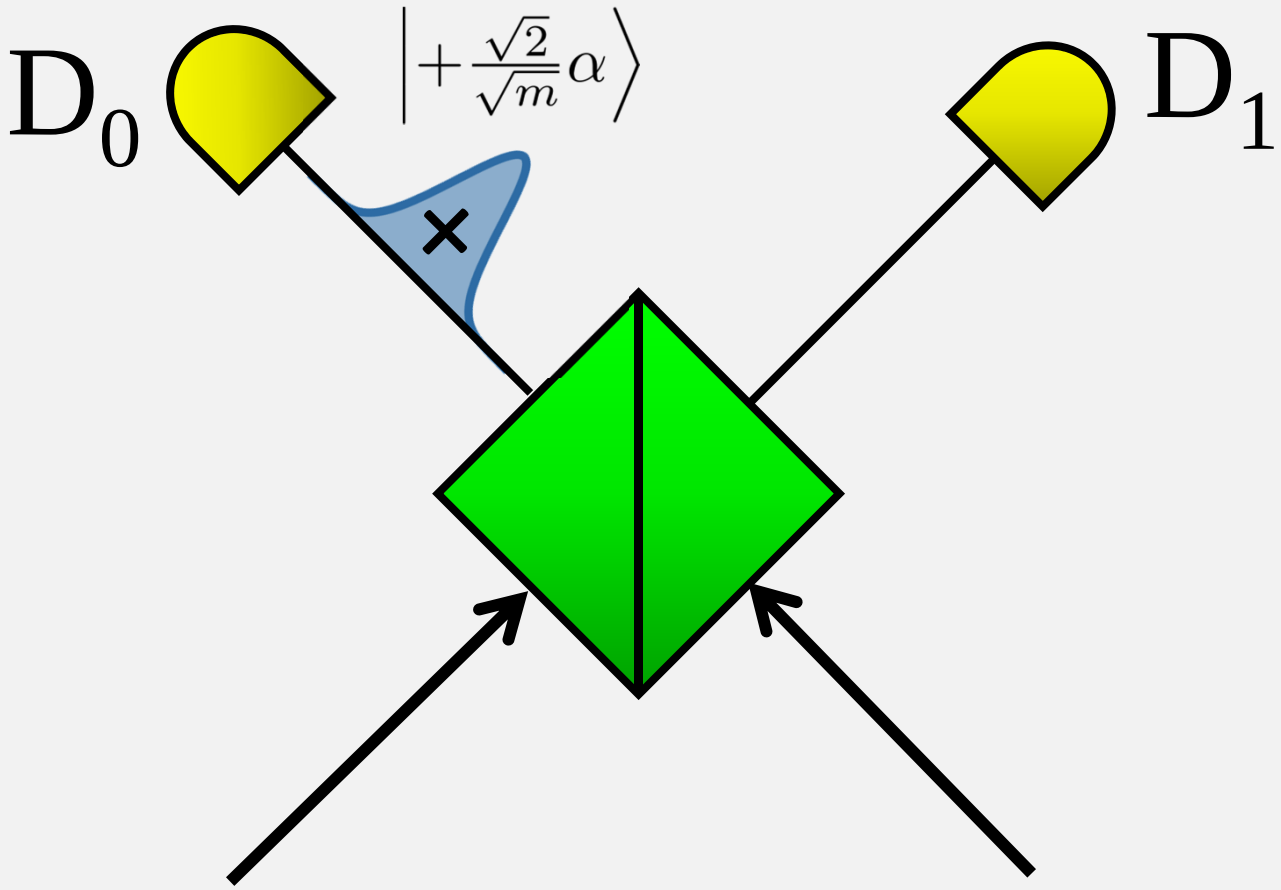


Arrazola, Lütkenhaus, *Phys. Rev. A,* **89**, 062305, (2014)
Xu, Arrazola et al. *arXiv:1503.05499*, Accepted by *Nature Commun.* (2015)

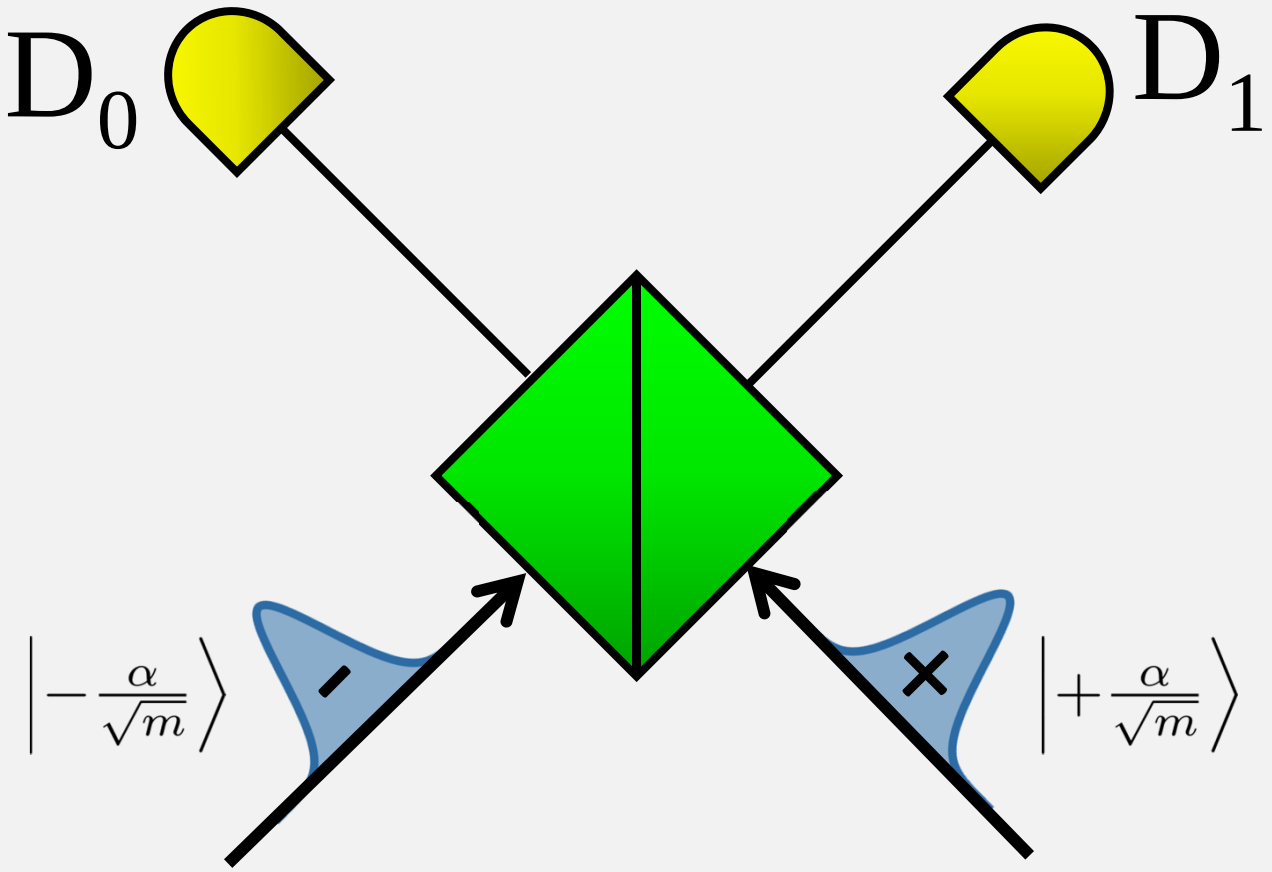
Interference results – Equal phases



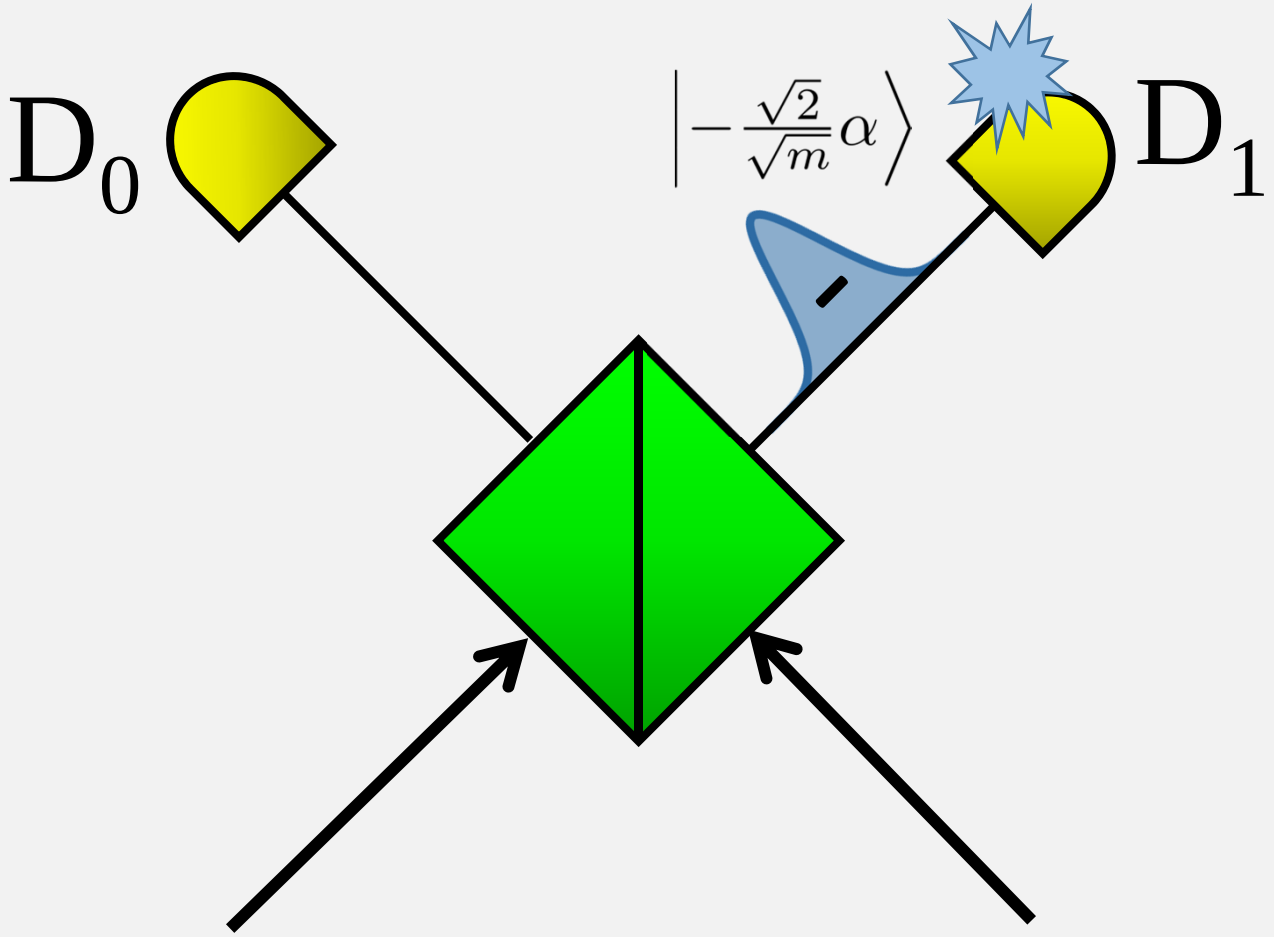
Interference results – Equal phases



Interference results – Different phases

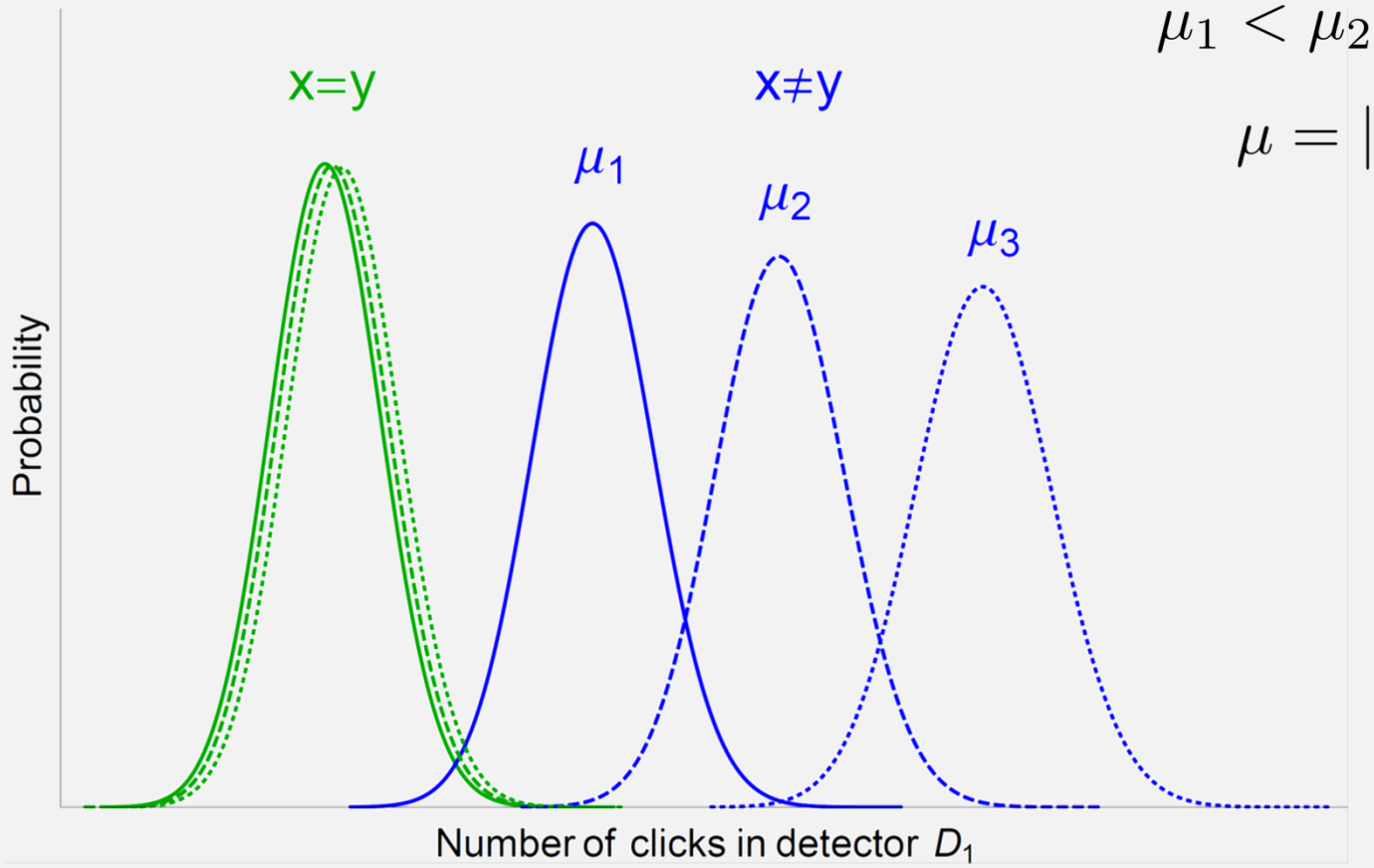


Interference results – Different phases



$$\Pr(\text{click}) = 1 - e^{-2\frac{|\alpha|^2}{m}} \approx 2\frac{|\alpha|^2}{m}$$

Referee's rule – checking D1



- Less clicks occur on D1 when X=Y than X≠Y.
- Both distributions become more distinguishable by increasing μ. We can achieve any probability of error by setting μ appropriately.

Xu, Arrazola et al. *arXiv:1503.05499*, Accepted by *Nature Commun.* (2015)

Experiment

Feihu Xu

Dept. of Electrical and Computer Engineering,
University of Toronto

Currently: Shapiro-Wong group, MIT

Major challenges to implementation

1. The initial coherent-state protocol suggested detectors with a very low dark count rate ~ 50 Hz



2. It requires good control and stable interference of very weak coherent states ~ 10^{-5} photons per pulse

3. Requires high visibility of a large number of laser pulses ~ 10^{8}

Can we relax the requirements and demonstrate the quantum fingerprinting protocol with commercial devices?



Arrazola, Lütkenhaus, *Phys. Rev. A,* **89***,* 062305, (2014)

Our Solution

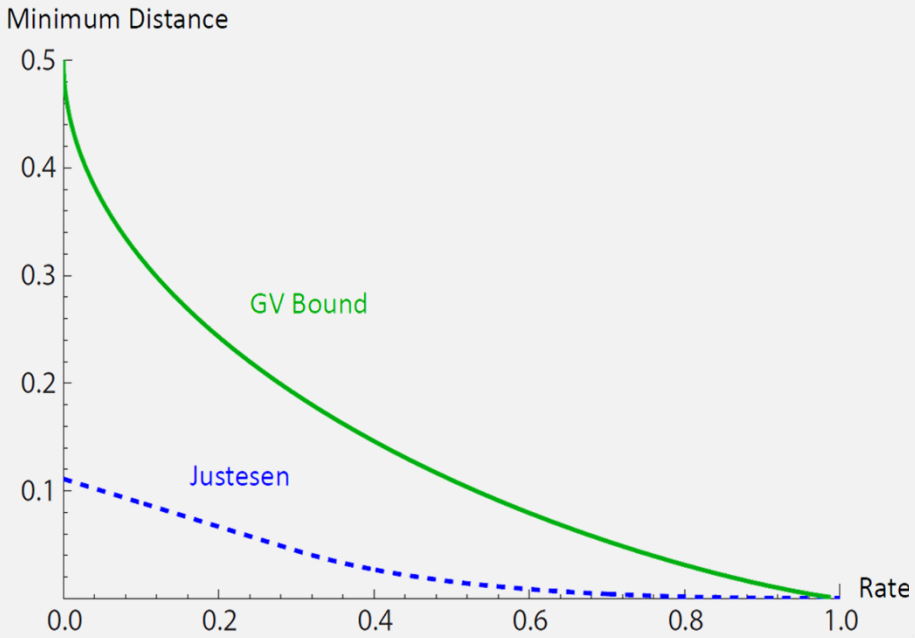
- Constructed a more efficient error correcting code to relax the demanding requirements for the devices

- Adopted a modified commercial plug&play system to perform the high-visibility interference for ultra-weak coherent states

Xu, Arrazola et al. *arXiv:1503.05499*, Accepted by *Nature Commun.* (2015)

Error Correcting Code

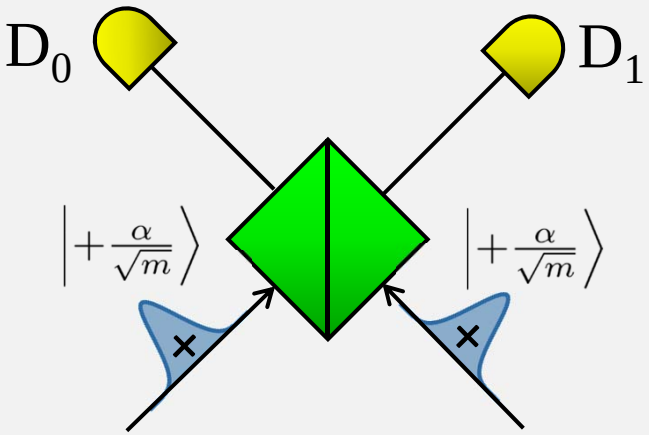
- Random Linear Codes (RLCs)
- Quasi-linear encoding complexity
- RLCs are easy to encode, but hard to decode
- Decoding is NOT required in quantum fingerprinting

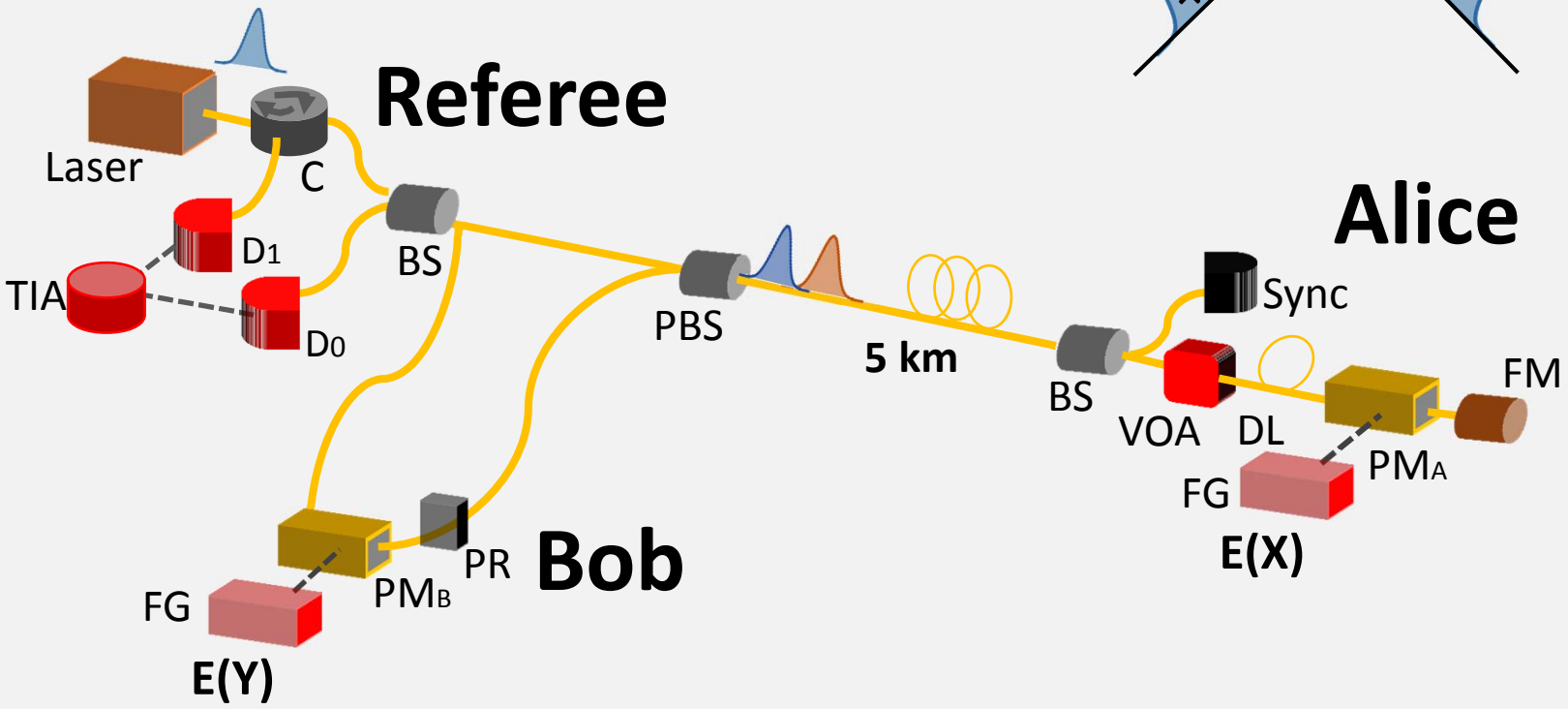


- ➢ Approaching Gilbert-Varshamov bound
- ➢ The minimum distance is <u>three times</u> larger than that of previous code, but with an epsilon failure probability added
- ➢ Relax detectors to a dark count rate ~ 5 KHz

Experimental setup

Modified commercial Plug&Play system (manufactured by ID Quantique).





The ‘Red’ devices indicate the modifications.

http://www.idquantique.com/

Additional challenges for P&P system

1. The detectors in Clavis2 have relatively low efficiency (~ 8%) and high dark counts (~ 20 KHz)

2. Commercial Clavis2 does not allow external input bit values for phase encoding

3. The low mean photon number (10^{-5} per pulse) is outside the normal QKD parameters (~ 0.1 per pulse)

Our modifications

1. External free-running InGaAs avalanche photodiodes (IDQ 220)



20% efficiency & 2~3 KHz dark counts
http://www.idquantique.com/

2. Two external function generators to modulate the phases according to codewords E(X) and E(Y)

3. Control the variable optical attenuator ~ 10^{-5} photons per ns pulse

➢ Precise synchronization, calibration and polarization control

➢ Transmit 100 Mbits over a 5 km fiber with a visibility = $(99 \pm 0.5)\%$

Real set up

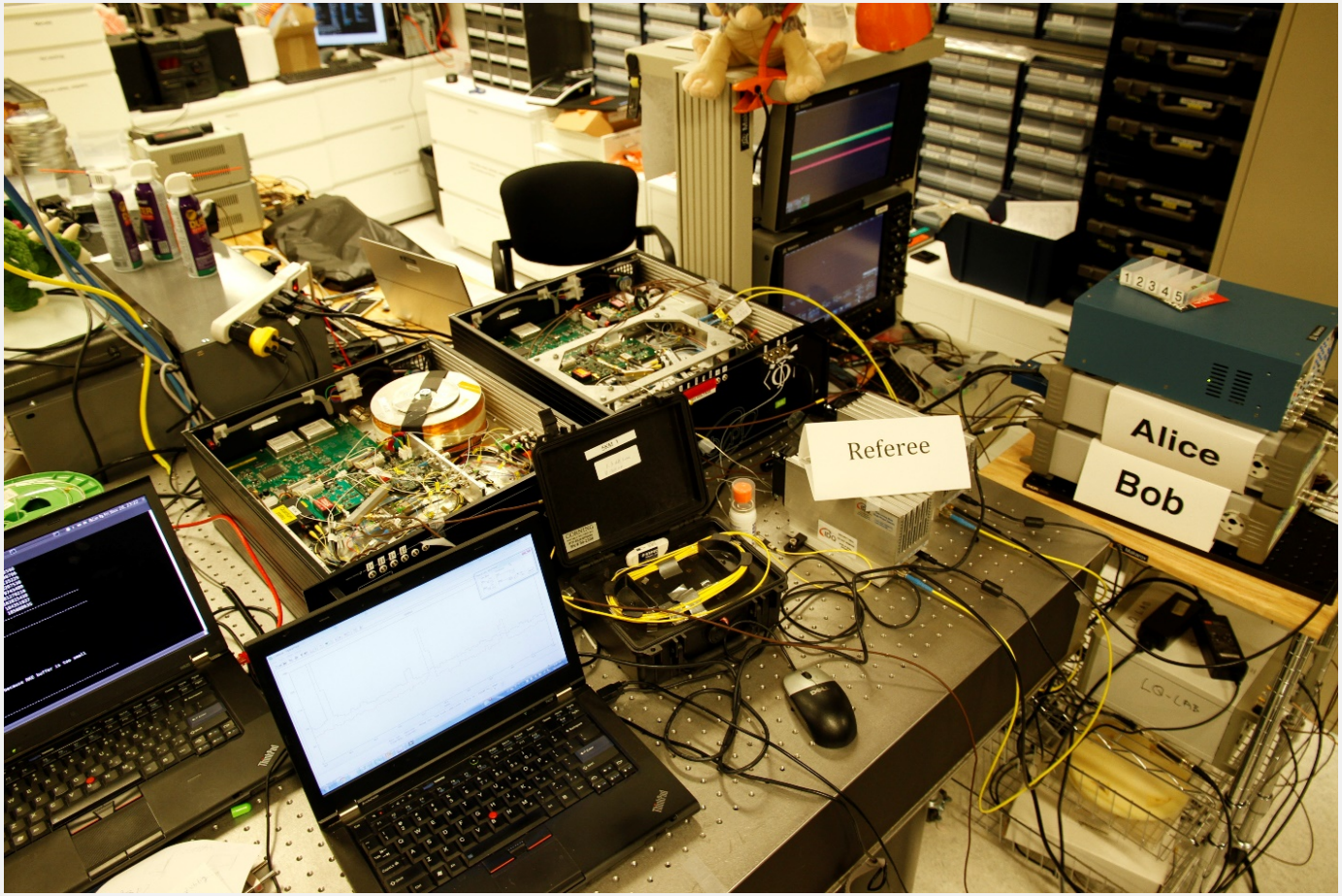
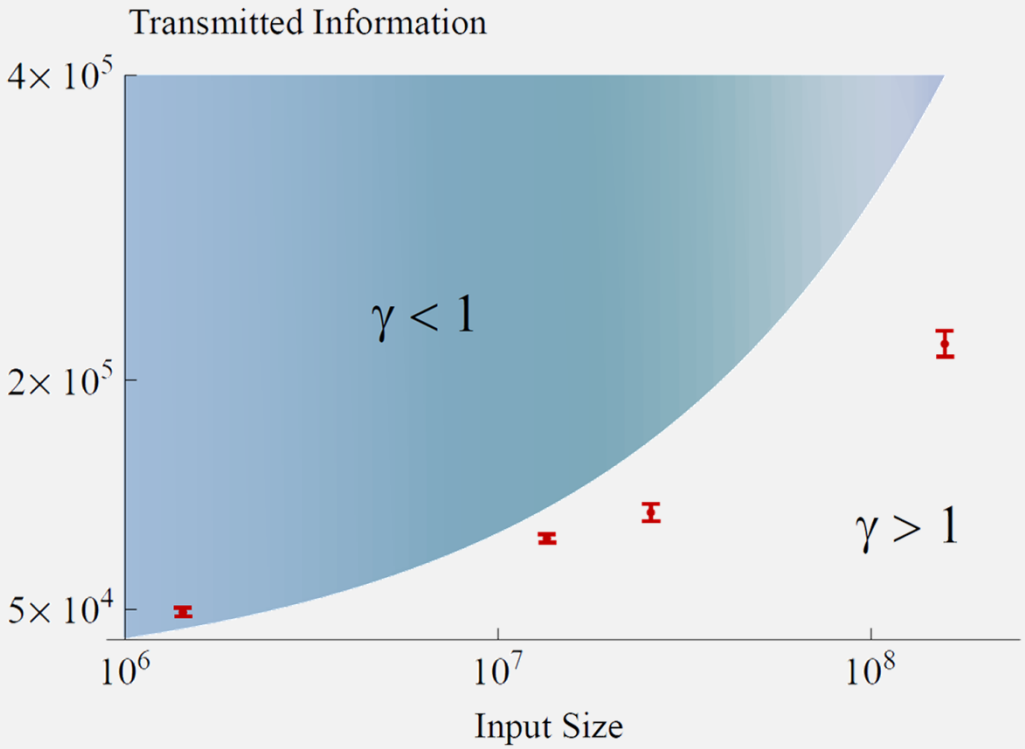


Photo credit: Vadim Makarov

Experimental results

Quantum advantage: $\gamma = \frac{C}{Q}$



For messages up to 100 Mbits, the information transmitted is 66% lower than the best known classical protocol

Summary

ARIGATO!

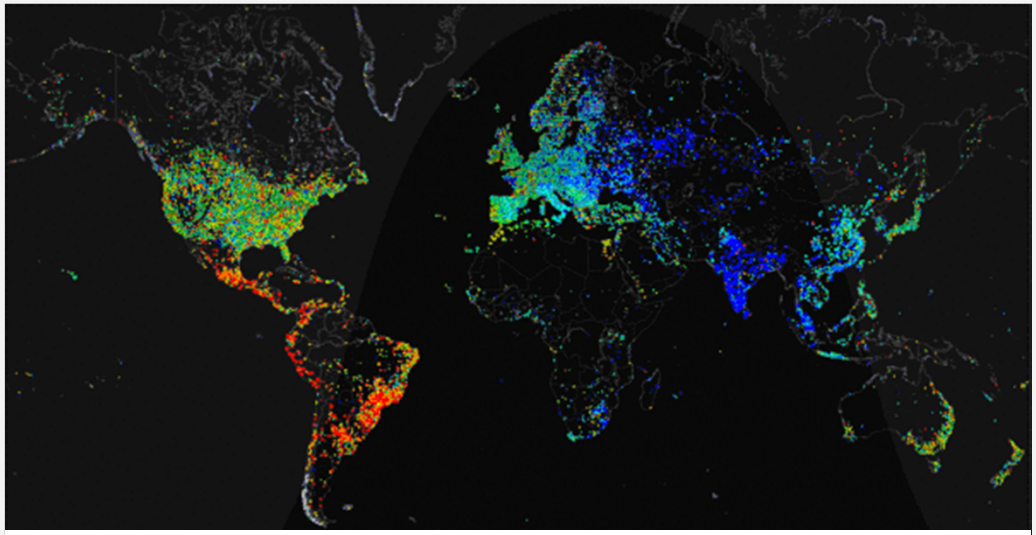
- ✓Propose a practical coherent-state protocol
- ✓Design a decoding rule and perform simulations to consider experimental imperfections
- ✓Develop a new ECC that enables the implementation with commercial devices
- ✓Construct a modified commercial plug&play system to demonstrate the protocol

- ➢The first proof of concept quantum fingerprinting system that transmits less information than the best known classical protocol

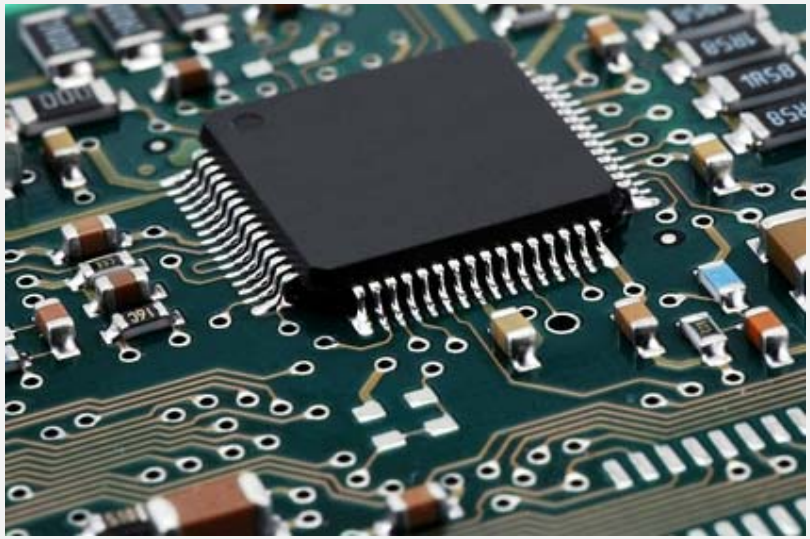
F. Xu *et al.* Accepted by *Nature Commun. arXiv:1503.05499* (2015)

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Communication complexity



World map of daily utilization of IPv4 addresses



Find the *minimum* amount of communication needed to solve distributed computational tasks.

- Fundamental Physics and CS
- Green communication and networks
- VLSI circuit design
- Data structure design

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Physicists Triumph at Guess My Number

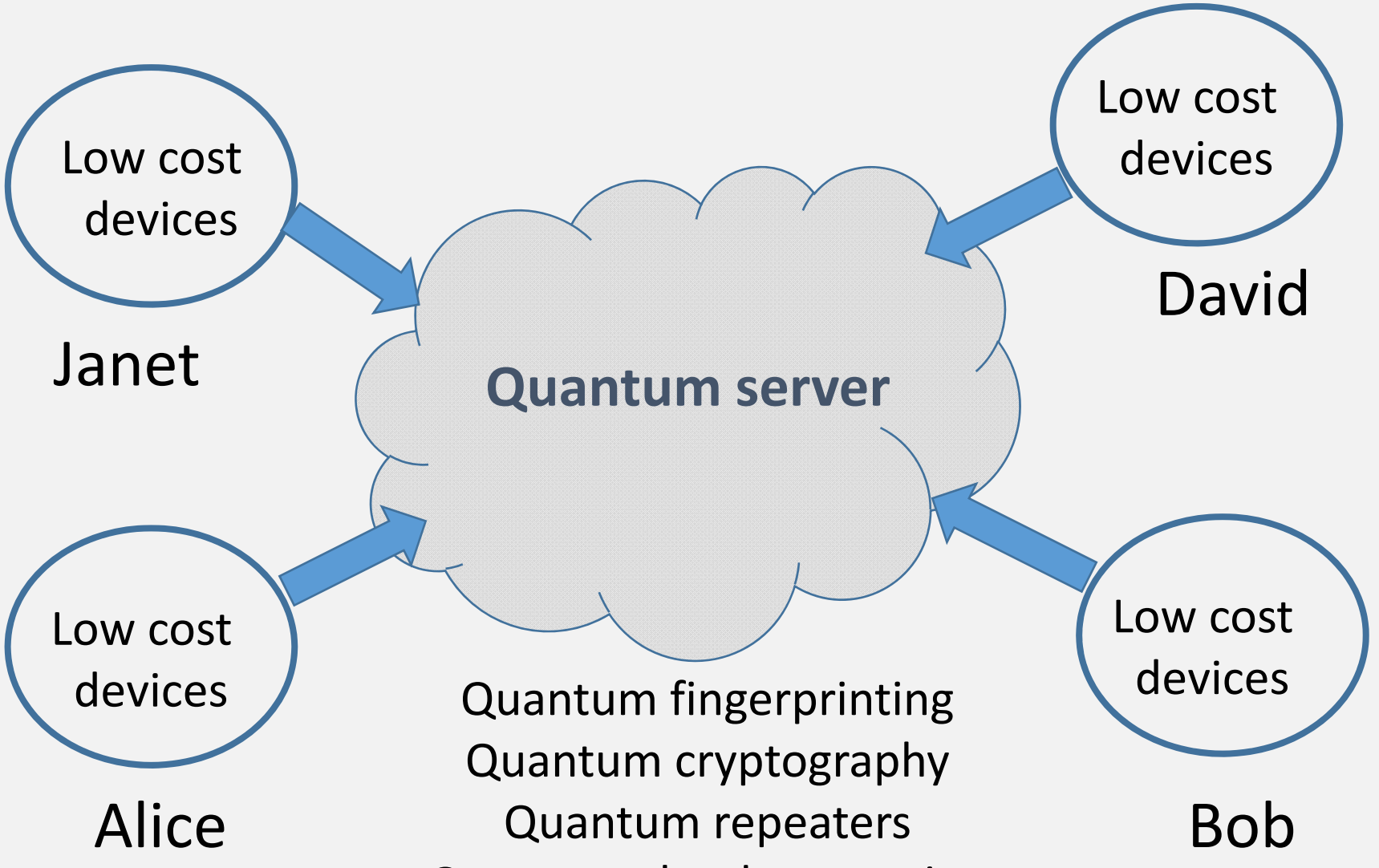
Andrew M. Steane and Wim van Dam

Back up – quantum?

- Non-orthogonality of weak coherent states (WCS)
- Quantum mechanical properties of single-photon detectors.

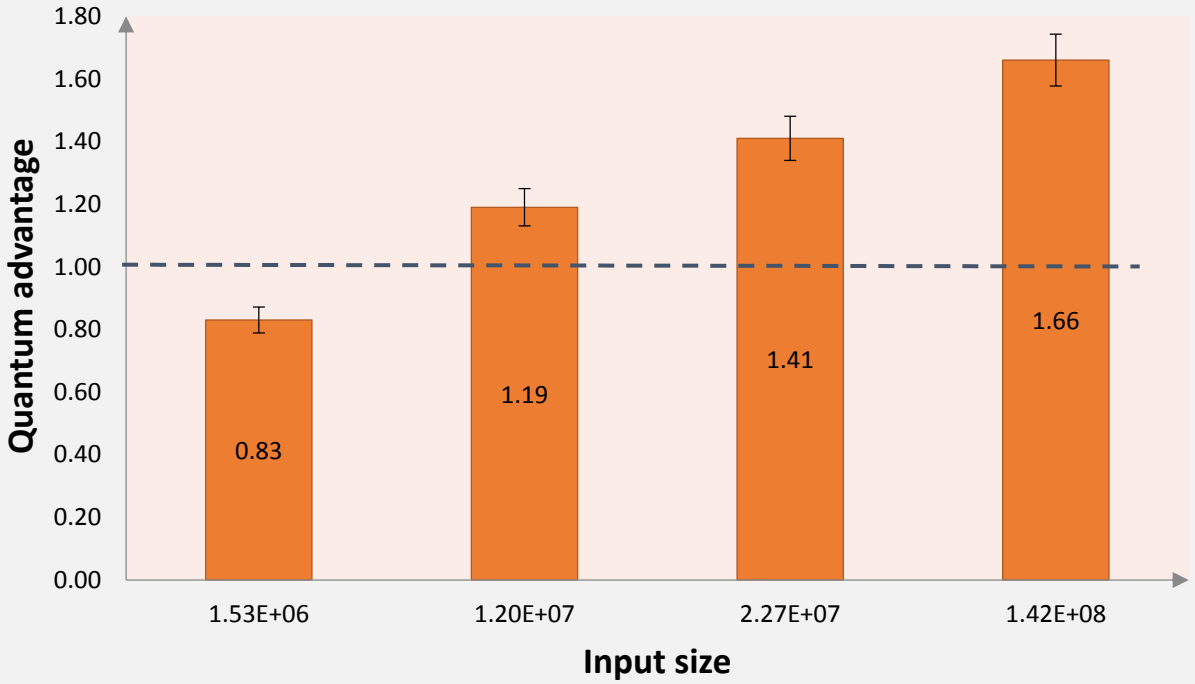
- The WCS have a very low mean photon number ~10^-5. This means that the two possible states that are sent in each mode (with a plus or minus phase) are highly non-orthogonal and difficult to distinguish.
- Very little information can be learnt by looking at each pulse. This is the reason why the transmitted information is very low – exponentially less than in the classical case.
- After the WCS interfere in the beam-splitter, a click in the single-photon detector unambiguously provides valuable information to the referee: she now knows whether the phases of the WCS are equal or not.
- The referee uses this information to determine whether the inputs to Alice and Bob are equal, even if very little information was sent to him. This unambiguous information is only possible because the detectors respond quantum mechanically to the incoming light field.

Outlook – Quantum Internet



Quantum fingerprinting
Quantum cryptography
Quantum repeaters
Quantum cloud computing
…

Experimental results



For messages up to 100 Mbits, the information transmitted is **66%** lower than the best known classical protocol.

Practical imperfections

- Taking into account practical errors
  - Detector efficiency (~20%), dark counts (~3 KHz)
  - Channel loss (~2 dB) and error (~1%)
  - Imperfect interference (visibility ~ 99%)
  - …

